

Where Do Vision-Language Models Preserve and Use Depth Information?

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Key Findings

- Metric depth is decodable at nearly every layer, beyond what image height and size alone can provide.
- But its causal effect is confined to an early LM layer range.

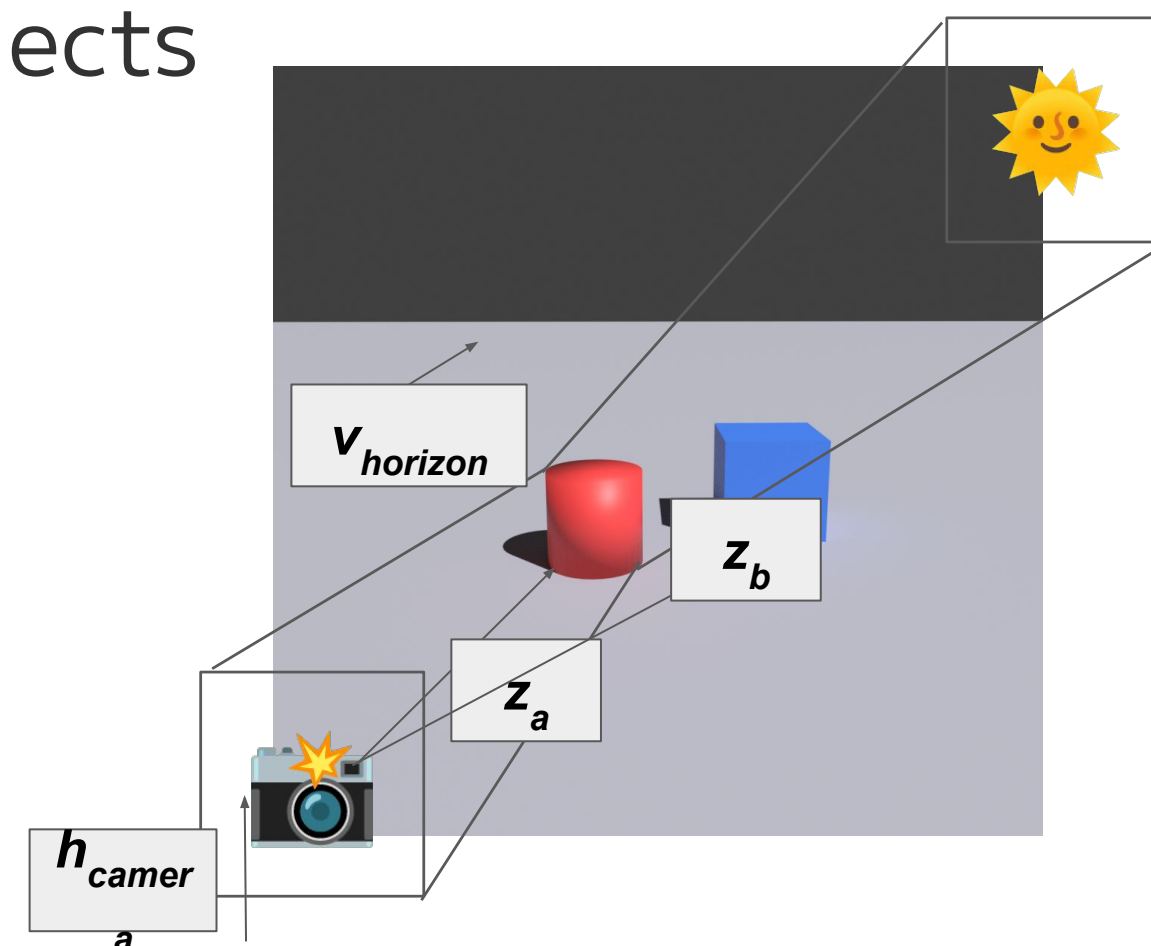
Motivation

- **Metric depth, distance in metres, is key to 3D understanding.**
 - For robotics and 3D reconstruction [Atar+, IROS'25; Yin+, ICCV'23].
- **Do VLMs internally represent metric depth and use it?**
 - We probe if depth is decodable beyond simple visual cues.
 - We intervene to test if it guides answers.

- 😊 Spatial relations are decodable and causally effective in VLMs. (cf. 2D relation [Kang+, ICLR'26], occlusion [Maheshwari+, arXiv'26], 3D topology [Wang & Gao, arXiv'26])
- 😞 Yet metric depth remains underexplored inside VLMs.
- 😞 Depth is strongly correlated with 2D cues. (e.g., Nearer objects appear lower and larger.)
- ➔ Prior work breaks that tie at render time [Min+, ECCV'26]; whereas we benchmark against these cues.

Task Definition

- We give the VLM an image with two objects at different depths z_A and z_B .
- We ask
 - **Q:** "Which one's centre is a shorter distance from the camera?"
 - **A:** A or B
- and record its answer.



Dataset Construction

- We generated 1,620 synthetic scene.
 - two objects on the ground plane.
 - known depth z , under controlled camera and lighting.
- On a ground plane, depth z and image position v are tied, and the relationship can be written as:

$$\frac{v - v_{\text{horizon}}}{\text{how far below the horizon (px)}} \propto \frac{f \cdot h_{\text{camera}}}{z} \quad \begin{matrix} f : \text{focal length (px)} \\ h_{\text{camera}} : \text{camera height (m)} \end{matrix}$$

metric depth (m)

Conditions

- To reduce bias, we generated data under 3 conditions (540 each)
 - same apparent size, depth differs
 - each object appears both near and far position
 - random sampling, no additional constraints

Construction Process

1. Sample a camera pose, height, pitch and yaw jittered.
2. Sample the pair's depth at a depth ratio. (depth range 2.95–6.65 m)
3. Render and record ground truth information. (depth z , mask and horizontal row)

Analysis Protocol

Probing

- **To analyze if the model represents depth via linear probing:**
 1. We pool the object's visual-token activations at each layer.
 2. We fit a linear model to predict its depth in metres.

$$\hat{z} = w^T h + b$$

activation vectors object's visual-tokens $\hat{z}$: predicted metric depth
 w : direction used to decode depth
 h : pooled object's visual-token activations
 b : bias

- Probe performance may reflect simple visual cues, not depth.
 - ➔ For comparison, we fit a linear model using five visual cues:
 - ① height below horizon
 - ② horizontal off-centre
 - ③ apparent height
 - ④ apparent area
 - ⑤ bbox fill

Intervention

- **To analyze if internal depth affects the model's choice:**
 1. We shift decoded depth using the probe model.
 - Probe direction defines the depth axis.
 - We shift the object state along it by α metres.

$$h_t \leftarrow h_t + \alpha \frac{w}{w^T w}, t \in \mathcal{T}_{\text{obj}}$$

$\mathcal{T}_{\text{obj}} = \{t_1, t_2, \dots, t_{|\mathcal{T}_{\text{obj}}|}\}$: set of the object's visual-token indices
 h_t : activation vector at visual token t

2. We read the A/B logits, how strongly the model favors each, and define the **margin** as their difference.

$$\text{margin} = \ell_{\text{target}}^{(\alpha)} - \ell_{\text{other}}^{(\alpha)}$$

$\ell_{\text{target}}^{(\alpha)}$: target-answer logit after an α -metre depth shift
 $\ell_{\text{other}}^{(\alpha)}$: other-answer logit after an α -metre depth shift

3. We measure belief gap Δ between farther and nearer shifts.

$$\Delta = \text{margin}_{+\alpha} - \text{margin}_{-\alpha}$$

pushed farther pushed nearer

➔ High belief gap $\Delta \Rightarrow$ The model uses internal depth

Results

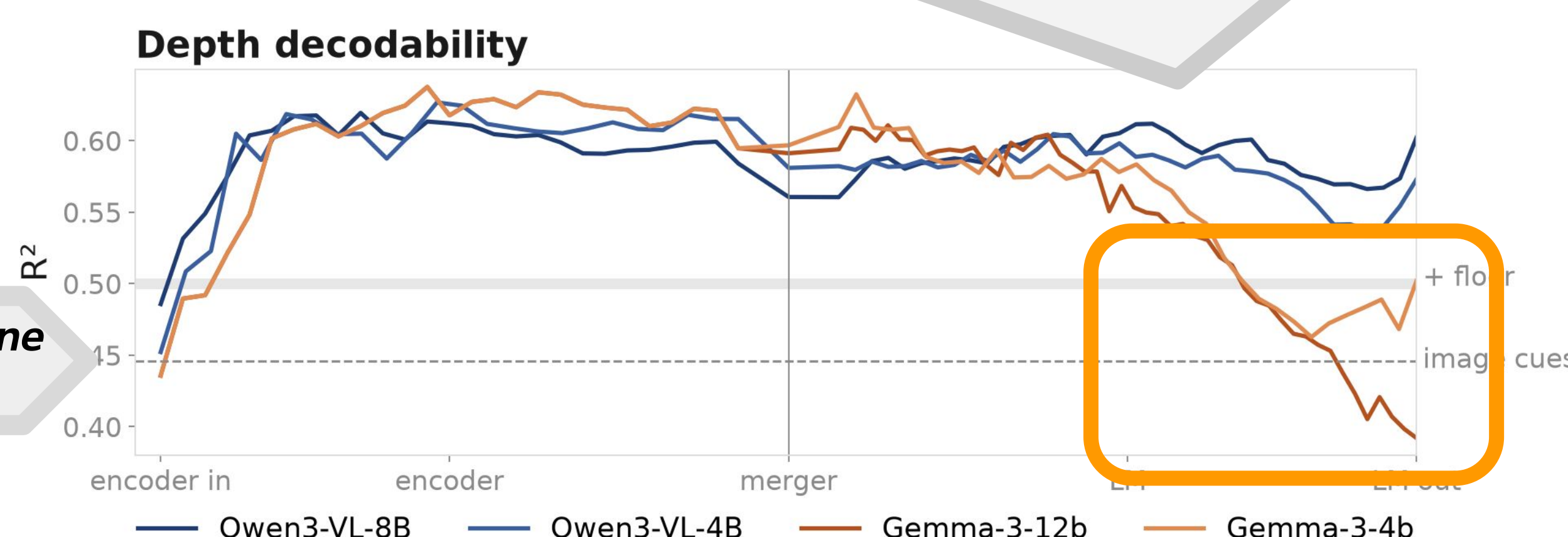
Synthetic data accuracy

model	acc.
Qwen3-VL-8B [Bai+, arXiv'25]	0.947
Qwen3-VL-4B [Bai+, arXiv'25]	0.863
Gemma-3-12b [Gemma Team, arXiv'25]	0.600
Gemma-3-4b [Gemma Team, arXiv'25]	0.515

lower task accuracy lose decodable metric depth in later LM layers

Probing

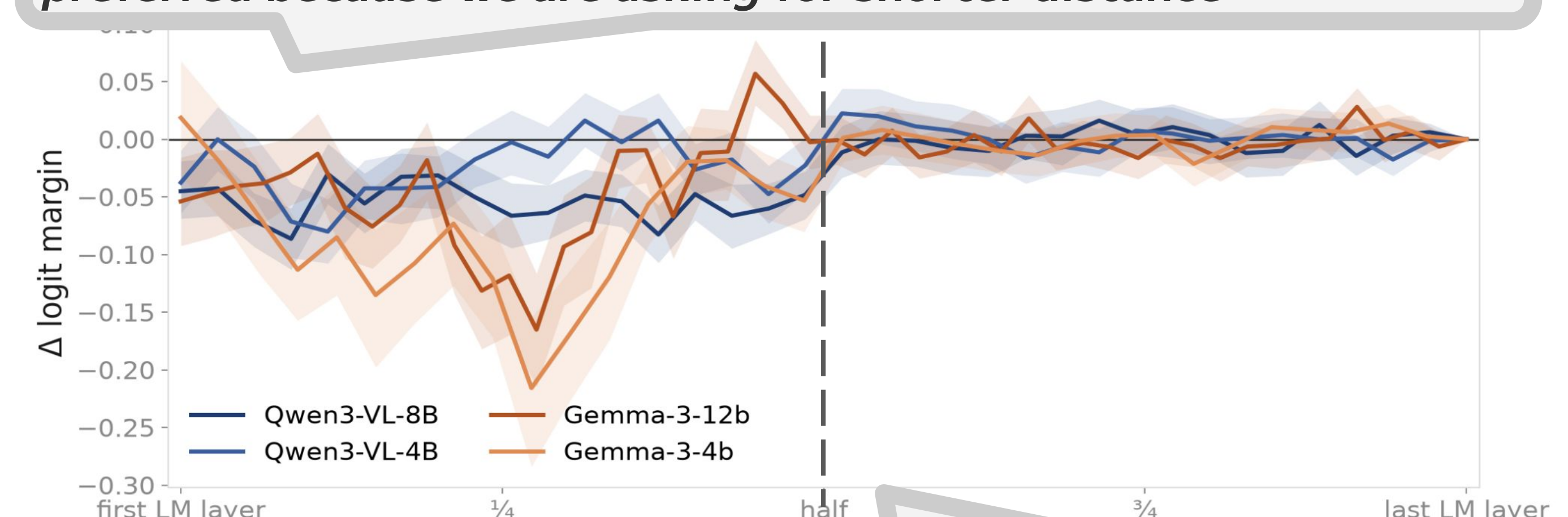
Most layers successfully decoded metric depth



Intervention

Each point is the result of intervention at that layer alone, and the y-axis shows how strongly the intervention affected the answer

The further from zero, the stronger the effect. Negative Δ is preferred because we are asking for shorter distance



Intervention in an early LM layer shifts the model's belief in the expected direction

Discussion & Future Work

- Early-LM Intervention selectively shifts belief, but does not show that depth is used for the final decision.
 - ➔ The intervention effect is too small to cross the decision margin and also appears without depth-dependent behavior.
- Metric depth cannot be fully separated from its 2D cues in the current dataset.
- The relationship between objects is not considered.